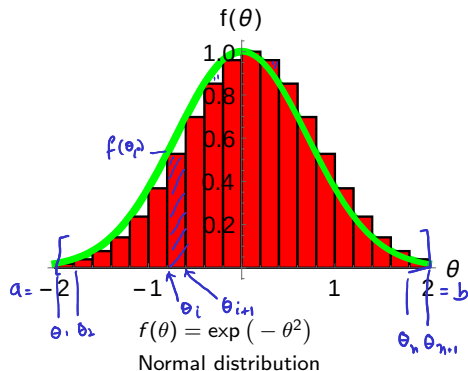




Riemann Int.

Partition into small intervals $\Delta\theta_i = \theta_{i+1} - \theta_i > 0$

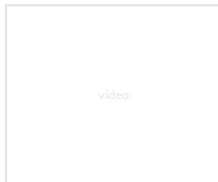
$$a = \theta_1 \leq \theta_2 \leq \theta_3 \leq \dots \leq \theta_{n+1} = b$$

$n = 20$

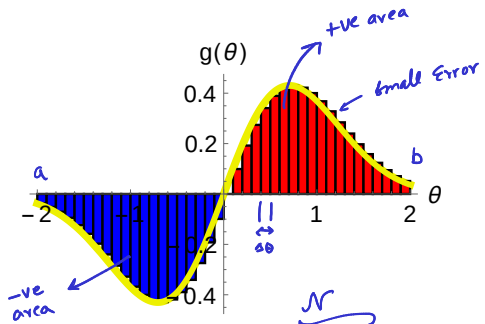
$$\sum_{i=1}^n \underbrace{f(\theta_i) \Delta\theta_i}_{\text{area of a red rectangle}} \approx \int_a^b f(\theta) d\theta$$

n very large
 $n \rightarrow \infty$
 $\Delta\theta \rightarrow d\theta$

Integration is the total area under the green curve.



Integration: Expectation values and Moments



$$g(\theta) = \theta f(\theta) = \theta \exp(-\theta^2)$$

The first moment (mean) $E(\Theta) = 0$

Partition into small intervals $\Delta\theta_i = \theta_{i+1} - \theta_i > 0$

$$a = \theta_1 \leq \theta_2 \leq \theta_3 \leq \dots \leq \theta_{n+1} = b$$

$n = 40$

$$E(\Theta) = \sum_{i=1}^n \theta_i \underbrace{f(\theta_i) \Delta\theta_i}_{p_i} \approx \int_a^b \theta f(\theta) d\theta$$

Integration is the total area under the **yellow** curve.

